

CSC 425 – Operating Systems and Architecture

Student Syllabus

Course Information

Credits: 3
Location: S&E 341
Day/Time: MWF 9:05am – 10:00am

Prerequisites: CSC 133 CSC 204

Instructor Information

Instructor: Mr. Scott Madeira
Office: S&E 343C
Office Hours: Wednesday 1:30pm – 3:30pm or
by Appointment

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Course Description

CSC425: Introduction to operating systems and computer architecture with systems software programming exercises utilizing the C language. Topics include: the kernel abstraction, process management, concurrency, synchronization, memory management and persistent storage. Fall semester.

Course Introduction

CSC425 is designed to introduce the student to computer operating systems, the functions they perform, how they are constructed and how they function with computer hardware. The course will consist of lectures on the theory and programming assignments to reinforce the learning.

Outcomes

Upon successful completion of this course a student is able to do the following:

1. Demonstrate an understanding of basic computer architecture.
2. Demonstrate an understanding of operating system components.
3. Demonstrate competency in writing multi-threaded programs.
4. Explain how information is passed between processes and threads in a Linux operating system.

Resources

- Laptop computer with code editor for doing lab exercises and project work
- Prescribed textbooks
- Geneva lab and email accounts
- Other resources as the class constructs them

Course Procedures and Requirements

Class time will consist of a combination of lectures on operating system and architecture topics and work periods to apply learning to programming projects. It is expected that the student will need to spend considerable time outside of class completing the programming projects.

The class is exam and project-based. Grading is based on individual performance but collaborating on everything but exams is encouraged. Students are expected to do their own work, however.

Attendance Policy

The instructor will follow the *Excused Absence Policy* listed in the College Catalog for all students. If a student is going to miss a class for a campus event, they should inform the instructor in **advance**.

In order to make-up missed work, students will have access to course materials which may include notes, PowerPoint slides, or other material naturally generated during the delivery of the lesson. *There will not be an option to participate in class via videoconference.*

It is the student's responsibility to communicate with the faculty member directly regarding their need to miss class for any reason. It is our expectation that a student will be truthful in communicating with faculty about class absences. Please remember that misrepresenting a class absence is a violation of the academic integrity policy and could result in a failing grade in the course or other penalties including dismissal from the college.

Students are expected to attend class. However, there is no penalty for missing class. The student's chances for success without coming to class are not very good.

Classroom Materials

Original graded papers, assignments and projects will be returned to the student. The instructor may keep copies of student work for purposes of instruction or as examples for future courses. Course quizzes and examinations (questions and answers) are the property of the Professor and may be retained by the instructor. Examinations and other graded work completed at the end of the semester that is not returned to the student will be saved for one year before being discarded so that the student can refer to them or retrieve them.

Learning and Physical Disabilities Policy

Geneva College values diversity and inclusion and recognizes disabilities as an aspect of diversity. Our shared goal is to create learning environments that are accessible, equitable, and inclusive; ultimately leading to the success of our students in and beyond college. Therefore, Geneva College complies with the ADA as well as Section 504 of the Rehabilitation Act, by affording reasonable accommodations to qualified students with disabilities. Any student with a disability who needs academic accommodations should contact Geneva's Student Success Center (SSC) to arrange a confidential appointment with the Director of the SSC before or during the first week of classes. (Some accommodations may take time to put into place, so it is advised to request this as early as possible.) Legally, no retroactive accommodations can be provided.

Accommodations for disabilities are available only as recommended by the SSC. For more details, visit Geneva's website, call 724.847.5005, or stop by the office. They are located on the second floor of the

Student Center. Students who have questions about their right to benefits or have grievances under these statutes should contact the Director of the SSC.

Geneva College also offers professional counseling services for students who are experiencing concerns with their overall mental health, either chronic or acute. These services are confidential and free of charge and are provided by Health and Counseling Services. Students can schedule an appointment through

MyGeneva: [https://my.geneva.edu/ICS/Student Information/Health and Counseling Services.jnz](https://my.geneva.edu/ICS/Student%20Information/Health%20and%20Counseling%20Services.jnz)

or contact Counseling Services at 724-847-4081.

Electronic Devices

While in the physical classroom, to maintain the focus in the classroom for all students and the professor on the lecture, please silence all cell phones, pagers, PDAs, laptops and any other form of electronic communication upon entering the classroom. Inform your friends and family that your phone will generally be off-limits during class time except for true emergencies. Unless explicitly authorized by the instructor, use of any such device during an exam for anything other than responding to a true emergency is strictly prohibited and may result in failure of the exam, at the discretion of the instructor. If an emergency occurs and you must use your phone, please leave the classroom to do so.

The Grading System

The grading scale is A, B, C, D, and F. Geneva includes pluses (+) and minuses (-) in the determination of the grade point average (GPA). Grade points are assigned for each semester hour of credit earned according to the following system:

Letter Grade	Percentage	GPA	Explanation
A	93.00% - 100.00%	4.0	Excellent
A-	90.00% - 92.99%	3.7	
B+	87.00% - 89.99%	3.3	
B	83.00% - 86.99%	3.0	Good
B-	80.00% - 82.99%	2.7	
C+	77.00% - 79.99%	2.3	
C	73.00% - 76.99%	2.0	Satisfactory
C-	70.00% - 72.99%	1.7	
D+	67.00% - 69.99%	1.3	
D	63.00% - 66.99%	1.0	Less than Satisfactory
D-	60.00% - 62.99%	0.7	
F	< 60.00%	0.0	No Credit Given

Course Evaluation

Criteria	Weight	Points
Exams (3)	45%	450 points
Project 0	5%	50 points
Project 1	10%	100 Points
Project 2	15%	150 points
Project 3	20%	200 points
Participation	5%	50 points
Total		1000 points

- Participation consists of coming to class prepared and engaging in the discussion as well as being an active participant on Discord and helping fellow students.

Policy on Academic Honesty

You may work together on homework assignments and programming projects, but you must complete your own work and turn in your own files. Do not submit a file completed by someone else or significantly like someone else's file. Programming is a very unique and individual activity. While you should all get to the same destination on the projects, each student will arrive there on very different paths. It will be very obvious if you are using someone else's code. If it is discovered that you are copying someone else's work (exam, homework, case assignments, project) you will receive a zero for that assignment. A second offense will result in failure for the entire class. Please spare all of us the hassle, headache and embarrassment that results from exercising this policy.

AI-based tools: We treat AI-based assistance, such as ChatGPT and Copilot, the same way we treat collaboration with other people: you are welcome to talk about your ideas and work with other people, both inside and outside the class, as well as with AI-based assistants.

However, all work you submit must be your own. You should never include in your assignment anything that was not written directly by you without proper citation (including quotation marks and in-line citation for direct quotes). Including anything you did not write in your assignment without proper citation will be treated as an academic dishonesty case. If you are unsure where the line is between collaborating with AI and copying from AI, we recommend the following heuristics:

Heuristic 1: Never hit "Copy" within your conversation with an AI assistant. You can copy your own work into your conversation, but do not copy anything from the conversation back into your assignment. Instead, use your interaction with the AI assistant as a learning experience, then let your assignment reflect your improved understanding.

Heuristic 2: Do not have your assignment and the AI agent open at the same time. Similar to above, use your conversation with the AI as a learning experience, then close the interaction down, open your assignment, and let your assignment reflect your revised knowledge. This heuristic includes avoiding using AI directly integrated into your composition environment: just as you should not let a classmate write content or code directly into your submission, so also you should avoid using tools that directly add content to your submission.

Deviating from these heuristics does not automatically qualify as academic dishonesty; however, following these heuristics essentially guarantees your collaboration will not cross the line into misconduct.

Turnitin Policy

Students agree that by taking this course all required papers may be subject to submission for textual similarity review to [Turnitin.com](https://turnitin.com) for the detection of plagiarism. All submitted papers will be included as source documents in the Turnitin reference database solely for the purpose of detecting plagiarism of such papers. Students must agree to the End User License Agreement (EULA) for Turnitin. Use of the Turnitin service is subject to the Usage Policy and Privacy Pledge posted on Turnitin.com.

Academic Grievance

For grievances related to grading and other course academic policies, the student should follow the published *Academic Grievance Policy* in the College Catalog.

Course Schedule/Outline

This schedule is tentative and will be altered as the instructor deems appropriate during the semester.

	CSC 425		Operating Systems and Architecture	Fall 2023
Week	Date	Chapter	Lecture	Homework / Notes
1	28-Aug		Course Introduction	Install IDE
	30-Aug	0	C - Basics	Install Virtual Machine
	1-Sep		C - Pointers	
2	4-Sep		C - Memory functions / Strings	
	6-Sep			
	8-Sep	1	Operating System Introduction	
3	11-Sep			Program 0 - Linked Lists
	13-Sep	2	Operating System Structures	
	15-Sep			
4	18-Sep	3	Processes	
	20-Sep			
	22-Sep		Project Work Day	
5	25-Sep			P1 - Sockets
	27-Sep	4	Threads	
	29-Sep			
6	2-Oct			
	4-Oct	5	Concurrency	
	6-Oct			
7	9-Oct			
	11-Oct		Exam Review	
	13-Oct		Exam #1 - Chapters 2, 3, 4, 5	
8	16-Oct		FALL BREAK - NO CLASS	
	18-Oct		Pthreads	
	20-Oct			
9	23-Oct	6	CPU Scheduling	
	25-Oct			
	27-Oct		Project Work Day	
10	30-Oct			P2 - Multi-threading
	1-Nov	8	Main Memory	
	3-Nov			
11	6-Nov			
	8-Nov	9	Virtual Memory	
	10-Nov			
12	13-Nov			
	15-Nov	12	File- System Implementation	
	17-Nov			
13	20-Nov		Exam #2 - Chapters 6, 8, 9, 12	
	22-Nov			
	24-Nov		THANKSGIVING BREAK - NO CLASS	
14	27-Nov			
	29-Nov	13	I/O Systems	
	1-Dec			P3 - Shared Memory
15	4-Dec			
	6-Dec	16	Virtual Machines	
	8-Dec			
16	11-Dec	17	Distributed Systems	
	15-Dec		FINAL EXAM 1:00pm - Chapters 16, 17	